

Q1. Solve the system

$$\left( \begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array} \right)$$

$$\rightarrow \left( \begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 0.5 & 0.5 & 1 \\ 0 & 2 & 1 & 5 \end{array} \right)$$

$$\rightarrow \left( \begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 0.5 & 0.5 & 1 \\ 0 & 0 & -1 & 1 \end{array} \right)$$

back-sub:  $z = -1$ ,  $0.5y - 0.5 = 1 \Rightarrow y = 3$ ,  $2x + 3 + 1 = 8 \Rightarrow x = 2$

$x = 2$ ,  $y = 3$ ,  $z = -1$

Q2. Rank and nullity

rref has 2 pivots, so rank  $A = 2$

nullity  $= 3 - 2 = 1$

null space:  $(-1, 2, 1)$

Q3. Subspace proof

$$W = \{ (x, y, z) : x + 2y - z = 0 \}$$

Need: closure under  $+$  and under scalar multiplication.

Take  $(1, 0, 1)$  and  $(0, 1, 2)$ , both in  $W$ .

Sum is  $(1, 1, 3)$ :  $1 + 2 - 3 = 0$ , so in  $W$ .

$3 \cdot (1, 0, 1) = (3, 0, 3)$ :  $3 + 0 - 3 = 0$ , in  $W$ .

So  $W$  is a subspace.

Q4. Inverse

$$[A|I] =$$

$$\left( \begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$\rightarrow \left( \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$\rightarrow \left( \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$